

Rotational Dynamics

- Relation between Linear and Angular parameters

- ① Relation between Linear displacement (s) and Angular displacement (θ)

$$s = r\theta$$

- ② Relation between linear velocity (v) and angular velocity (ω)

$$v = r\omega$$

- ③ Relation between linear acceleration and angular acceleration (α)

$$a = r\alpha$$

- Equations of motion for rotational motion

① $\omega = \omega_0 + \alpha t$

② $\theta = \omega_0 t + \frac{1}{2} \alpha t^2$

③ $\omega^2 = \omega_0^2 + 2\alpha\theta$

- Relation between angular velocity (ω), time period (T) and frequency (f)

$$\omega = \frac{2\pi}{T} = 2\pi f$$

• Moment of Inertia (I) or Rotational Inertia

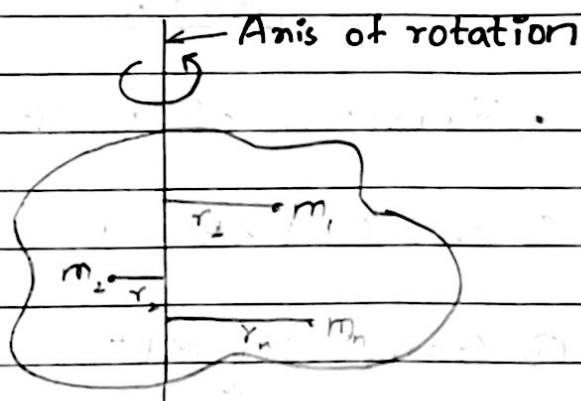
→ The moment of Inertia of a rigid body about an axis of rotation is defined as the tendency of the body to resist the change of its rotational motion.

Mathematically,

moment of inertia of a rigid body is defined as the sum of product of masses of its constituent particles and square of their distances from the axis of rotation.

$$\text{i.e. } I = m_1 r_1^2 + m_2 r_2^2 + \dots + m_n r_n^2$$

$$\therefore I = \sum_{i=1}^n m_i r_i^2$$



→ Moment of inertia depends upon

- (i) Mass of Body
- (ii) Distribution of mass relative to axis of rotation.
- (iii) Shape of body

(iv) Position and orientation of axis of rotation.

Q Why Spokes are fitted in bicycle wheel?

Ans Spokes ~~are~~ fitted in bicycle wheel increases the moment of inertia of the wheel and hence the cycle runs smoothly.

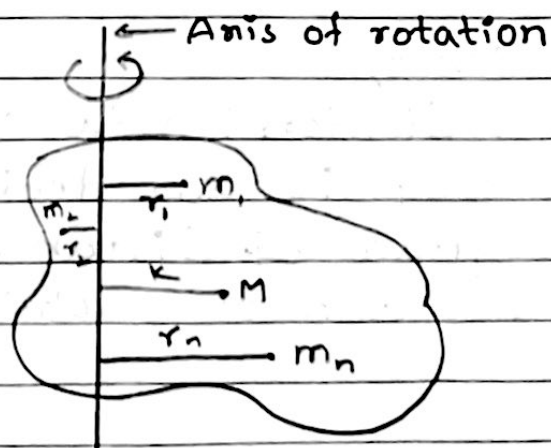
• Radius of gyration (K)

→ The radius of gyration of a rigid body is defined as the perpendicular distance between the axis of rotation and the point at which whole mass of the body is supposed to be concentrated such that its moment of inertia remains same.

The moment of inertia of a body of mass ' M ' in terms of ~~radius of gyration~~ of radius of gyration is given by

$$I = MK^2 \longrightarrow \text{①}$$

Consider a rigid body consists of n particles of masses $m_1, m_2, \dots, m_n$ at the perpendicular distances $r_1, r_2, \dots, r_n$ respectively



from the axis of rotation as shown in fig.

Then, moment of inertia of the body is also given by

$$I = m_1 r_1^2 + m_2 r_2^2 + \dots + m_n r_n^2 \rightarrow (2)$$

From (1) & (2)

$$MK^2 = m_1 r_1^2 + m_2 r_2^2 + \dots + m_n r_n^2$$

For simplicity,

$$\text{Let, } m_1 = m_2 = \dots = m$$

then,

$$MK^2 = m r_1^2 + m r_2^2 + \dots + m r_n^2$$

$$\text{or, } MK^2 = m (r_1^2 + r_2^2 + \dots + r_n^2)$$

Since,

$$M = nm \quad \text{So,}$$

$$nmK^2 = m (r_1^2 + r_2^2 + \dots + r_n^2)$$

$$\text{or, } K^2 = \frac{r_1^2 + r_2^2 + \dots + r_n^2}{n}$$

$$\therefore K = \sqrt{\frac{r_1^2 + r_2^2 + \dots + r_n^2}{n}} \rightarrow (3)$$

Hence,

~~Radius~~ Radius of gyration of a rigid body is also defined as the root mean square of distance of its particles from the axis of rotation.

- Radius of gyration depends upon:

① Distribution of mass relative to axis of rotation.

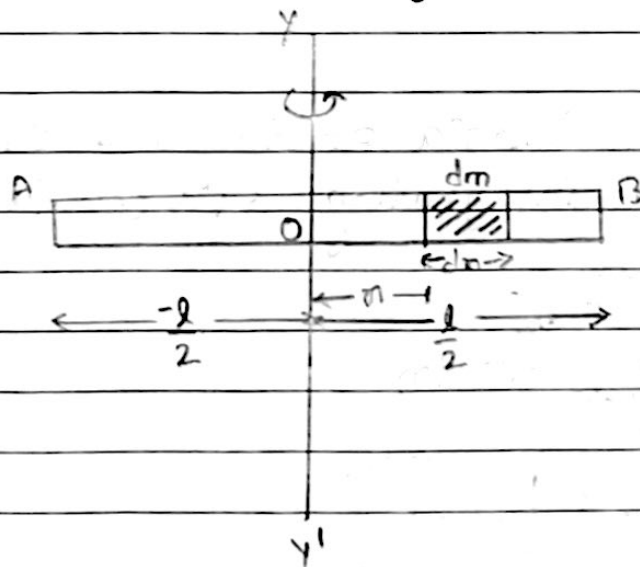
② Shape of body

③ Position and orientation of axis of rotation.

Note: Radius of gyration doesn't depend upon mass of body.

✓ ~~Moment~~ of Inertia of a thin uniform rod

① About an axis through its centre



Consider a thin uniform rod AB of length l and mass M is rotating about an axis YY' passing through its centre

o as shown in figure.

Let us take a small strip of length dm and mass dm at distance ' x ' from the axis of rotation. Then moment of inertia of the strip is given by

$$dI = dm \times x^2 \quad \text{--- (1)}$$

Here,

$$\text{Mass per unit length of rod} = \frac{M}{l}$$

$$\therefore \text{mass of length } dm \text{ of rod i.e. } dm = \frac{M}{l} dm$$

Then,

From eqn (1)

$$dI = \frac{M}{l} dm \times x^2 \quad \text{--- (2)}$$

Now,

moment of inertia of the rod can be obtained by integrating eqn (2) from $x = -\frac{l}{2}$ to $x = \frac{l}{2}$

$$\begin{aligned} \therefore I &= \frac{M}{l} \int_{-l/2}^{l/2} x^2 dx \\ &= \frac{M}{l} \times \left[\frac{x^3}{3} \right]_{-l/2}^{l/2} \\ &= \frac{M}{3l} \left[\left(\frac{l}{2} \right)^3 - \left(-\frac{l}{2} \right)^3 \right] \\ &= \frac{M}{3l} \left[\frac{l^3}{8} - \left(-\frac{l^3}{8} \right) \right] \end{aligned}$$

$$= \frac{M}{3l} \left[\frac{2l^3}{84} \right]$$

$$= \frac{Ml^2}{12} \longrightarrow \textcircled{3}$$

Note. If K be the radius of gyration of the rod then

$$I = MK^2 \longrightarrow \textcircled{4}$$

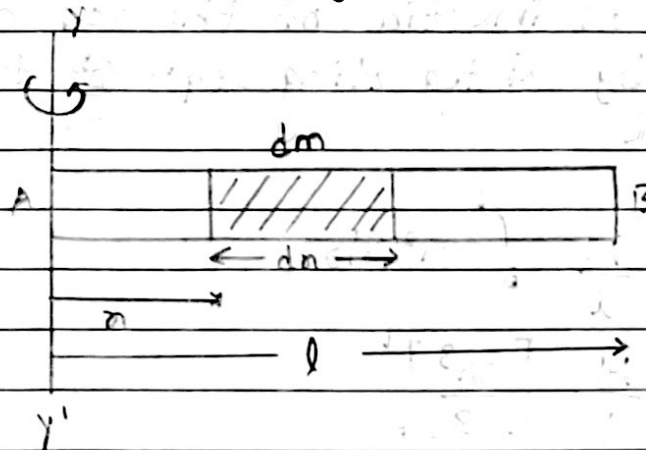
From $\textcircled{3}$ & $\textcircled{4}$

$$MK^2 = \frac{Ml^2}{12}$$

$$\text{or, } K^2 = \frac{l^2}{12}$$

$$\therefore K = \frac{l}{\sqrt{12}} \longrightarrow \textcircled{5}$$

② About an axis through its one end



Consider a thin uniform rod AB of length l and mass M is rotating about an axis YY' passing through its one end as shown in figure.

Let us take a small strip of length dx and mass dm at distance ' x ' from the axis of rotation. Then moment of inertia of the strip is given by

$$dI = dm \times x^2 \rightarrow (1)$$

Here,

$$\text{Mass per unit length of rod} = \frac{M}{l}$$

$$\therefore \text{mass of length } dx \text{ of rod} \\ \text{i.e. } dm = \frac{M}{l} dx$$

Then,

from eqn (1)

$$dI = \frac{M}{l} dx \times x^2$$

Now,

moment of inertia of the rod can be obtained by Integrating eqn (2) from $x=0$ to $x=l$

$$\therefore I = \frac{M}{l} \int_0^l x^2 dx$$

$$= \frac{M}{l} \left[\frac{x^3}{3} \right]_0^l$$

$$= \frac{M}{3l} \times [l^3 - 0]$$

$$= \frac{M}{3l} \times l^3$$

$$= \frac{Ml^2}{3} \rightarrow (3)$$

~~Also~~ Note: If k be the radius of gyration of the rod then,

$$I = Mk^2 \rightarrow (4)$$

From (3) & (4)

$$Mk^2 = \frac{ML^2}{3}$$

$$k^2 = \frac{L^2}{3}$$

$$\therefore k = \frac{L}{\sqrt{3}} \rightarrow (5)$$

• Torque (τ) or moment of force

↳ The turning effect of force on a body is called Torque.

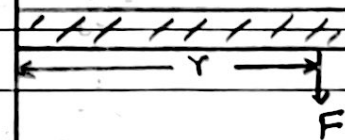
It is given by the vector product of applied force ($\vec{F}$) and distance ($\vec{r}$) of force from the axis of rotation.

$$\text{i.e. } \tau = \vec{r} \times \vec{F} \\ = rF \sin \theta \hat{n}$$

In magnitude

$$\tau = rF \sin \theta$$

← Axis of rotation



If $\theta = 90^\circ$ then

$$\tau = rF$$

• Relation between torque and moment of inertia

→ Consider a rigid body is rotating about an axis YY' with uniform ~~accel~~ ~~rotation~~ angular acceleration α . Suppose the body consists of n particles of masses $m_1, m_2, \dots, m_n$ at the perpendicular distances $r_1, r_2, \dots, r_n$ respectively from the axis of rotation.



Now,

Magnitude of torque acting on the particle of mass m_1 is given by

$$\begin{aligned} T_1 &= r_1 \times F_1 \\ &= r_1 \times m_1 a_1 \\ &= r_1 \times m_1 \times r_1 \alpha \\ &= m_1 r_1^2 \alpha \end{aligned}$$

Similarly, $T_2 = m_2 r_2^2 \alpha$

$$T_3 = m_3 r_3^2 \alpha$$

$\vdots$

$$T_n = m_n r_n^2 \alpha$$

Then, total torque acting on the body is given by

$$\begin{aligned} T &= T_1 + T_2 + T_3 + \dots + T_n \\ &= m_1 r_1^2 \alpha + m_2 r_2^2 \alpha + m_3 r_3^2 \alpha + \dots + m_n r_n^2 \alpha \\ &= \alpha (m_1 r_1^2 + m_2 r_2^2 + m_3 r_3^2 + \dots + m_n r_n^2) \end{aligned}$$

$$= \left(\sum_{i=1}^n m_i r_i^2 \right) \alpha$$

$$= I \alpha$$

Where $I = \sum_{i=1}^n m_i r_i^2$ is moment of inertia of the body.

$$\therefore \tau = I \alpha$$

This gives relation between torque and moment of inertia.

• Angular momentum ($\vec{L}$)

→ The moment of linear momentum of a body is called Angular Momentum.

$$\therefore \vec{L} = \vec{r} \times \vec{p}$$

Where, $\vec{p}$ = Linear momentum

$\vec{r}$ = distance of body from the axis of rotation.

$$\therefore \vec{L} = r p \sin \theta \hat{n}$$

In magnitude,

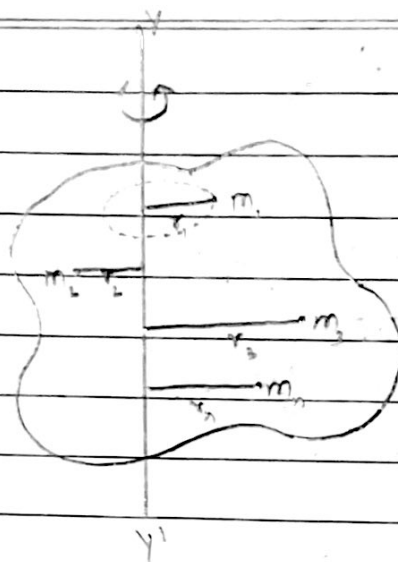
$$L = r p \sin \theta$$

When $\theta = 90^\circ$ then,

$$L = r p$$

• Relation between Angular momentum and Moment of Inertia

→ Consider a rigid body is rotating ab-



out an axis YY' with angular velocity ω . Suppose the body consists of n particles of masses $m_1, m_2, \dots, m_n$ at the perpendicular distances $r_1, r_2, r_3, \dots, r_n$ respectively from the axis of rotation. Now,

Angular momentum of the particle of mass m_1 is given by,

$$\begin{aligned} L_1 &= r_1 \times p_1 \\ &= r_1 \times m_1 \times v_1 \\ &= r_1 \times m_1 \times r_1 \omega \\ &= m_1 r_1^2 \omega \end{aligned}$$

Similarly,

$$L_2 = m_2 r_2^2 \omega$$

$$L_3 = m_3 r_3^2 \omega$$

⋮

$$L_n = m_n r_n^2 \omega$$

Then,

Total Angular momentum of the body is given by,

$$\begin{aligned}
 L &= L_1 + L_2 + L_3 + \dots + L_n \\
 &= m_1 r_1^2 \omega + m_2 r_2^2 \omega + m_3 r_3^2 \omega + \dots + m_n r_n^2 \omega \\
 &= (m_1 r_1^2 + m_2 r_2^2 + m_3 r_3^2 + \dots + m_n r_n^2) \omega \\
 &= \left(\sum_{i=1}^n m_i r_i^2 \right) \omega \\
 &= I \omega
 \end{aligned}$$

Where, $I = \sum_{i=1}^n m_i r_i^2$ is the moment

of inertia of the body.

$$\boxed{\therefore L = I \omega}$$

This gives relation between angular momentum and moment of inertia of the body.

- Relation between Torque and angular momentum.

→ Since $L = I \omega$

Differentiating both sides with respect to time 't', we have,

$$\frac{dL}{dt} = \frac{d(I \omega)}{dt}$$

$$\text{or, } \frac{dL}{dt} = I \frac{d\omega}{dt}$$

$$\text{or, } \frac{dL}{dt} = I \alpha \quad \text{Where } \alpha = \frac{d\omega}{dt} \text{ is Angular acceleration}$$

$$\text{or, } \frac{dL}{dt} = \tau$$

Where, $\tau = I\alpha$ is the torque.

$$\therefore \tau = \frac{dL}{dt}$$

Hence, Torque is defined as the ~~same~~ rate of change of angular momentum of a body with time.

- Principle of conservation of angular momentum.

Statement: It states that, "If no external torque is applied on a system then, total angular momentum of the system remains constant."

i.e. When $\tau = 0$ then $L = \text{constant}$

$$I\omega = \text{constant}$$

$$\text{i.e. } I_1\omega_1 = I_2\omega_2$$

Proof

Since,

Torque = Rate of change of angular momentum.

$$\therefore \tau = \frac{dL}{dt}$$

if, External torque is not applied on a system.

i.e. $\tau = 0$ Then,

$$\frac{dL}{dt} = 0$$

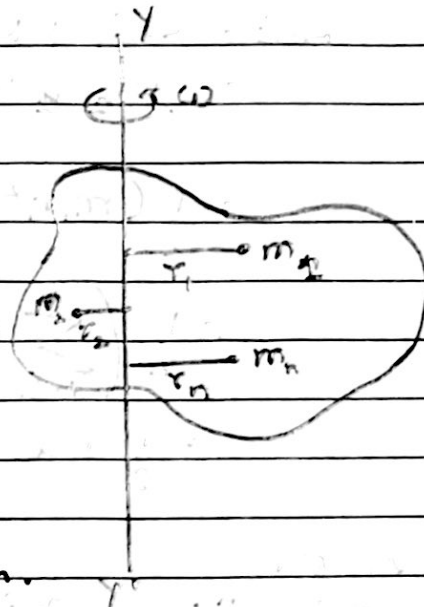
Integrating both sides, we get.

$$L = \text{constant}$$

$$\therefore I\omega = \text{constant}$$

• Rotational Kinetic Energy

Consider a rigid body is rotating about an axis YY' with angular velocity ω . Suppose the body consists of n particles of masses $m_1, m_2, \dots, m_n$ at the perpendicular distances $r_1, r_2, \dots, r_n$ respectively from the axis of rotation, as shown in figure.



Now, Kinetic Energy (K.E) of the particle of mass m_1 is given by

$$K.E_1 = \frac{1}{2} m_1 v_1^2$$

$$= \frac{1}{2} m_1 (r_1 \omega)^2$$

$$= \frac{1}{2} m_1 r_1^2 \omega^2$$

Similarly,

$$K.E_2 = \frac{1}{2} m_2 r_2^2 \omega^2$$

$$K.E_3 = \frac{1}{2} m_3 r_3^2 \omega^2$$

$$K.E_n = \frac{1}{2} m_n r_n^2 \omega^2$$

Then, total K.E of the body is given by

$$K.E = K.E_1 + K.E_2 + \dots + K.E_n$$

$$= \frac{1}{2} m_1 r_1^2 \omega^2 + \frac{1}{2} m_2 r_2^2 \omega^2 + \dots + \frac{1}{2} m_n r_n^2 \omega^2$$

$$= \frac{1}{2} (m_1 r_1^2 + m_2 r_2^2 + \dots + m_n r_n^2) \omega^2$$

$$= \frac{1}{2} \left(\sum_{i=1}^n m_i r_i^2 \right) \omega^2$$

$$= \frac{1}{2} I \omega^2 \rightarrow (1)$$

Where,

$I = \sum_{i=1}^n m_i r_i^2$ is the moment of inertia of body.

Since, $L = I\omega$

$$\therefore \omega = \frac{L}{I}$$

Then, from (1),

$$K.E = \frac{1}{2} I \times \left(\frac{L}{I} \right)^2$$

$$= \frac{1}{2} I \times \frac{L^2}{I^2}$$

$$= \frac{1}{2} \frac{L^2}{I}$$

$$\therefore K.E = \frac{L^2}{2I} \rightarrow (2)$$

Equations ① and ② are the required expression of rotational kinetic energy.

• Kinetic energy of rolling body

A rolling body possesses both the translational motion and rotational motion. So, kinetic energy of the rolling body is given by the sum of translational kinetic energy and rotational kinetic energy.

$$\therefore \text{K.E of rolling K.E} = \text{Translational K.E} + \text{Rotational K.E} \\ = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 \rightarrow \text{①}$$

Since, $I = mk^2$ $k \rightarrow$ radius of gyration.
and $\omega = \frac{v}{R}$ $R \rightarrow$ radius of body.

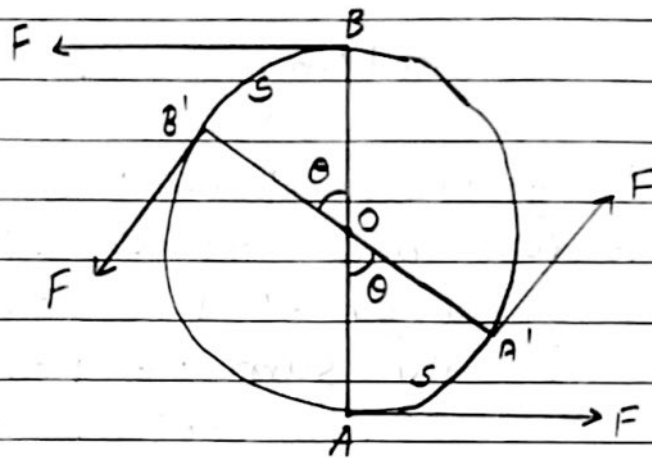
Then,

$$\text{rolling K.E} = \frac{1}{2}mv^2 + \frac{1}{2}mk^2 \times \frac{v^2}{R^2} \\ = \frac{1}{2}mv^2 \left(1 + \frac{k^2}{R^2} \right) \rightarrow \text{②}$$

• Work done by Couple

↳ Two equal and opposite forces acting on a

body such that their lines of action donot coincide are called a couple.



Consider a couple each of magnitude F acting on a wheel of radius (r) rotates the wheel by an angle as shown in figure.

Now,

Work done by the force (F) acting at 'A' is given by

$$W_A = FS$$

Where $S = AA'$ is linear displacement.

And, work done by the force (F) acting at B is given by,

$$W_B = FS$$

Where, $S = BB'$ is linear displacement.

Then,

Work done by couple is given by

$$W = W_A + W_B$$

$$= FS + FS$$

$$= 2FS$$

Since,

$$S = r\theta$$

$$\begin{aligned}\therefore W &= 2Fr\theta \\ &= (F \times 2r) \theta \\ &= T\theta\end{aligned}$$

Where, $T = F \times 2r$ is the torque due to couple.

$$\therefore W = T\theta$$

• Rotational power

→ The rate of doing work is called power.

$$\therefore P = \frac{dW}{dt}$$

$$= T \frac{d\theta}{dt}$$

$$= T\omega$$

Where $\omega = \frac{d\theta}{dt}$ is angular velocity.

$$\therefore P = T\omega$$

Numerical

Linear motion	Rotational motion	Relation
Linear motion	Rotational motion	Relation
s	θ	$s = r\theta$
v	ω	$v = r\omega$
a	α	$a = r\alpha$
m	I	$I = mr^2$
F	τ	$\tau = r \times F$
p	L	$L = r \times p$

Linear motion	Rotational motion
1) $v = u + at$	$\omega = \omega_0 + \alpha t$
2) $s = ut + \frac{1}{2} at^2$	$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$
3) $v^2 = u^2 + 2as$	$\omega^2 = \omega_0^2 + 2\alpha\theta$
4) $F = ma$	$\tau = I\alpha$
5) $p = mv$	$L = I\omega$
6) $F = \frac{dp}{dt}$	$\tau = \frac{dL}{dt}$
7) $K.E = \frac{1}{2} mv^2 = \frac{p^2}{2m}$	$K.E = \frac{1}{2} I\omega^2 = \frac{L^2}{2I}$
8) $W = Fs$	$W = \tau\theta$
9) $\text{Power}(p) = \frac{W}{t} = Fv$	$\text{power}(p) = \frac{W}{t} = \tau\omega$

1) $\theta = 2\pi n \Rightarrow n \rightarrow \text{no. of rotations}$

2) $I = Mk^2 \Rightarrow k \rightarrow \text{radius of gyration}$

3) $k = \sqrt{\frac{x_1^2 + x_2^2 + \dots + x_n^2}{n}}$

④ For rod

i) $I = \frac{ML^2}{12}$ (Axis through centre)

ii) $I = \frac{ML^2}{3}$ (Axis through one end)

⑤ Principle of Conservation of Angular momentum

$$L = \text{const}$$

$$\therefore I\omega = \text{constant}$$

$$\text{i.e. } I_1\omega_1 = I_2\omega_2$$

⑥ $\omega = 2\pi f = \frac{2\pi}{T}$

⑦ K.E of rolling body = $\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$

Question:

② A constant torque of 500 Nm turns a wheel about its centre. The moment of inertia about this axis is 100 kgm². Find the Angular Velocity gained in 4 second and kinetic energy gained after 20 revolutions.

→ Soln,

$$T = 500 \text{ Nm}$$

$$I = 100 \text{ kgm}^2$$

$$\omega = ?$$

$$T = 4 \text{ s}$$

$$\omega_0 = 0$$

$$K.E_0 = 0$$

$$K.E = ? \text{ When } n = 20$$

We know that

$$\tau = I \alpha$$

$$\Rightarrow 500 = 100 \alpha$$

$$\Rightarrow \alpha = \frac{500}{100}$$

$$\therefore \alpha = 5 \text{ rad/s}^2$$

then,

$$\omega = \omega_0 + \alpha t$$

$$= 0 + 5 \times 4$$

$$= 20 \text{ rad/s}$$

Again,

$$\theta = 2\pi n$$

$$\Rightarrow \theta = 2 \times \frac{22}{7} \times 20$$

$$= 40\pi \text{ rad}$$

Now,

$$\omega^2 = \omega_0^2 + 2\alpha\theta$$

$$\Rightarrow \omega^2 = 0 + 2 \times 5 \times 40\pi$$

$$\omega^2 = 400\pi$$

Then,

$$K.E = \frac{1}{2} I \omega^2$$

$$= \frac{1}{2} \times 100 \times 400\pi$$

$$= 20000\pi \text{ J}$$

$$= 62831.85307 \text{ J}$$

③ An electric Fan is turned off and its angular velocity decreases uniformly from 500 rev/min to 200 rev/min in 4 sec.

a) Find the angular acceleration and number of revolution made by the motor in 4 sec.

b) How many more seconds are required for the fan to come to rest if angular acceleration remains constant?

Soln $f_0 = 500 \text{ rev/min} = \frac{25}{3} \text{ rev/s}.$

$$\therefore \omega_0 = 2\pi f_0 = 2\pi \times \frac{25}{3} \times = \frac{50}{3} \pi \text{ rad/s}$$

$$f = 200 \text{ rev/min} = \frac{200}{60} = \frac{10}{3} \text{ rev/s}$$

$$\therefore \omega = 2\pi f = 2\pi \times \frac{10}{3} = \frac{20}{3} \pi \text{ rad/s}$$

$$t = 4 \text{ sec.}$$

a) $\alpha = ?$ and $n = ?$

Now,

$$\alpha = \frac{\omega - \omega_0}{t} = \frac{\frac{20}{3} \pi - \frac{50}{3} \pi}{4}$$

$$= \frac{-10\pi}{2}$$

$$= -7.85 \text{ rad/s}^2$$

Again,

$$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\begin{aligned}\theta &= \frac{50\text{ N}}{3} \times 4 + \frac{1}{2} \times \left(-\frac{5\text{ N}}{2}\right) \times 4^2 \\ &= \frac{200\text{ N}}{3} + (-20\text{ N}) \\ &= \frac{200\text{ N} - 60\text{ N}}{3} \\ &= \frac{140\text{ N}}{3} = 146.607 \text{ rad.}\end{aligned}$$

Then,

$$\begin{aligned}\theta &= 2\pi n \\ \Rightarrow \frac{140\text{ N}}{3} &= 2\pi n \\ \Rightarrow n &= \frac{70}{3} \text{ rev}\end{aligned}$$

$$\therefore n = 23.33 \text{ rev.}$$

- ① A constant torque of 500 Nm turns a wheel which has moment of inertia 20 kgm² about its center. Find the angular velocity gained in 2 second and the kinetic energy gained.

→ Soln,

$$T = 500 \text{ Nm}$$

$$I = 20 \text{ kgm}^2$$

$$\omega_0 = 0, \text{ ~~at~~ }$$

$$\omega = ? \text{ at } t = 2 \text{ sec}$$

$$K.E = ? \text{ When } t = 2 \text{ sec}$$

We know that,

$$T = I\alpha$$

$$500 = 20\alpha$$

$$\Rightarrow \alpha = \frac{500}{20}$$

$$\therefore \alpha = 25 \text{ rad/s}^2$$

then,

$$\begin{aligned}\omega &= \omega_0 + \alpha t \\ &= 0 + 25 \times 2 \\ &= 50 \text{ rad/s}\end{aligned}$$

Again,

~~$$\omega = \omega_0 + \alpha t$$~~

~~$$\omega = \omega_0 + \alpha t$$~~

$$K.E = \frac{1}{2} I \omega^2$$

$$= \frac{1}{2} \times 20 \times 50^2$$

$$= 10 \times 2500$$

$$= 25000 \text{ J}$$

- ④ A wheel starts from rest and accelerates with constant angular acceleration to an angular velocity of 15 rev/s in 10 seconds. Calculate the angular acceleration and angle which the wheel has rotated at the end of 2 second.

→ Soln,

$$\omega_0 = 0$$

$$f = 15 \text{ rev/s}$$

$$\omega = 2\pi f = 2\pi \times 15 = 30\pi \text{ rad/s when } t = 10 \text{ s}$$

$$\alpha = ? , \theta = ?$$

We know that,

$$\alpha = \frac{\omega - \omega_0}{t}$$

$$= \frac{30\pi - 0}{10}$$

$$= 3\pi = 9.42 \text{ rad/s}^2$$

Again,

$$\theta = \cancel{\omega_0 t} + \frac{1}{2} \alpha t^2$$

$$= \frac{1}{2} \times 9.42 \times 2^2$$

$$= 18.84 \text{ rad}$$

$$\therefore \theta = 18.84 \text{ rad}$$

3.b)

↳ Soln

$$\omega_{\text{new}} = 0$$

$$\omega_{\text{new}} = \frac{20}{3} \text{ rad/s}$$

$$t_{\text{entra}} = ?$$

$$\alpha = -7.85 \text{ rad/s}^2$$

Now,

$$\alpha = \frac{\omega_{\text{new}} - \omega_{\text{new}}}{t_{\text{entra}}}$$

$$-7.85 = \frac{0 - \frac{20}{3}}{t_{\text{entra}}}$$

$$\Rightarrow t_{\text{entra}} = \frac{+\frac{20}{3}}{+7.85}$$

$$\therefore t_{\text{entra}} \approx 2.66 \text{ sec}$$

⑤ Speed of a body spinning about an axis increases from rest to $100 \text{ rev. min}^{-1}$ in 5 sec. if a constant torque of 5 Nm is applied.

i) Calculate the moment of inertia of body about the axis.

ii) If external torque is then removed and body comes to rest in 100 s due to friction, calculate frictional torque.

→ Soln,

$$f_0 = 0, \omega_0 = 0$$

$$f = 100 \text{ rev/min} = \frac{100}{60} = \frac{5}{3} \text{ rev/s}$$

$$\omega = 2\pi f = 2\pi \times \frac{5}{3} \text{ rad/s} = \frac{10\pi}{3} \text{ rad/s}$$

$$T = 5 \text{ sec}$$

$$\tau = 5 \text{ Nm}$$

i) Soln,

$$\alpha = \frac{\omega - \omega_0}{T} = \frac{\frac{10\pi}{3} - 0}{5}$$

$$= 2.09 \text{ rad/s}^2$$

Now,

$$\tau = I \alpha$$

$$\Rightarrow 5 = I \times 2.09$$

$$\Rightarrow I = \frac{5}{2.09}$$

$$\Rightarrow I = 2.38 \text{ kgm}^2$$

ii) When $T = 100 \text{ s}$

$$\alpha' = \frac{\omega - \omega_0}{T}$$
$$= -\frac{\left(\frac{10}{3} \text{ rad} - 0\right)}{100}$$
$$= -0.1047$$

Now,

$$\tau' = I \alpha'$$
$$= 2.38 \times -0.10$$
$$= -0.24 \text{ Nm}$$

① A ballet dancer spins with 2.4 rev/s with her arms outstretched when the moment of inertia about the axis of rotation is I . With her arms folded, the moment of inertia about same axis becomes $0.6I$. Calculate

i) the new rate of spin.

ii) ~~the~~ ratio of her initial kinetic energy to the final kinetic energy of rotation.

↳ Soln,

$$f_1 = 2.4 \text{ rev/s}$$

$$\omega_1 = 2\pi \times 2.4 = 4.8\pi \text{ rad/s}$$

i) $I_1 \omega_1 = I_2 \omega_2$

$\Rightarrow I_1 \times 2.4 = 0.6 \times I_1 \times \omega_2$

$\Rightarrow \omega_2 = 4 \text{ rev/sec}$

ii) Soln,

$$\frac{K_1}{K_2} = \frac{\frac{1}{2} I_1 \omega_1^2}{\frac{1}{2} I_2 \omega_2^2}$$

$$= \frac{I_1 (2.4)^2}{0.6 \times I_2 \times (4)^2} \quad (\because \omega = 2\pi f)$$

$$= \frac{3}{5}$$

- ⑦ A disc of moment of inertia $5 \times 10^{-4} \text{ kgm}^2$ is rotating freely about the axis through its center at 40 rpm. Calculate new revolution per minute if some mass of 0.02 kg is dropped gently into the disc 0.08 m from the axis.

→ Soln,

$$I_1 = 5 \times 10^{-4} \text{ kgm}^2$$

$$f_1 = 40 \text{ rpm} = \frac{40}{60} = \frac{2}{3} \text{ rev/s}$$

$$m = 0.02 \text{ kg}$$

$$r = 0.08 \text{ m}$$

$$f_2 = ?$$

We know that

$$I_2 = I_1 + I_{\text{wan}}$$

$$= 5 \times 10^{-4} + mr^2$$

$$= 5 \times 10^{-4} + 0.02 \times (0.08)^2$$

$$= 6.28 \times 10^{-4} \text{ kgm}^2$$

Then,

$$I_1 \omega_1 = I_2 \omega_2$$

$$I_1 f_1 = I_2 f_2 \quad (\because \omega = 2\pi f)$$

$$5 \times 10^4 \times 40 = 6.28 \times 10^4 \times f_2$$

$$\Rightarrow f_2 = \frac{200}{6.28}$$

$$\therefore f_2 = 31.84 \text{ rev/min}$$

• Conceptual (short) Questions.

1) Define moment of inertia. State the factors on which it depends.

→ The tendency of the body to resist the change of its rotational motion is called moment of inertia.

It depends upon:

i) Mass of the body
ii) distribution of mass relative to axis of rotation.

iii) shape of the body

iv) position and orientation of axis of rotation.

2) Define radius of gyration. state the factor on which it depends.

→ The distance from the axis of rotation to the hypothetical point inside the body where whole mass of the body is supposed to be concentrated in such a way that the moment of inertia of the body about axis and moment of inertia of the point mass are equal.

It depends upon:

i) shape of the body

ii) distribution of mass relative to axis of rotation

iii) position and orientation of axis of rotation.

3) Explain why spokes are fitted in the cycle wheel?

→ The cycle wheel is so constructed so as to increase its moment of inertia with minimum possible mass. The greater the moment of inertia, the greater is the opposition to any change in uniform rotational motion. As a result, the cycle runs smoother and steadier. If the cycle wheel had no spokes, the cycle would be driven in jerks and hence unsafe.

4) Can you distinguish a raw egg and a hard boiled egg by spinning each one on the table? Explain.

→ Yes, we can distinguish them by spinning them.

- Hard boiled egg

- When spun, it behaves as a single solid body. It will spin rapidly and stop quickly when touched.

- Raw egg

- The interior is liquid. When you spin it, the liquid inside has inertia and continues to move or slosh around even when the shell is slowed. Consequently, it spins slower, feels "wobbly" and continues to rotate slightly after we touch it to stop it.

5) State the law of conservation of angular momentum. Give two illustrations of this law.

→ Law of conservation of angular momentum states that, "If the net external torque acting on a system is zero, the total angular momentum of the system remains constant.

$$(L = I\omega = \text{constant})$$

• Illustrations

① Ballet dancer

- A dancer spinning on ice brings their arms inward to decrease their moment of inertia, which causes their angular velocity to increase.

② Diver

- A diver tucks their body into a ball while in mid-air to increase the speed of their somersaults before extending again to enter the water.

6) A ballet dancer stretches her hands when she wants to come to rest. Why?

→ When a ballet dancer wants to slow down or come to rest, they stretch their hands outward. This action increases their moment of inertia (I). Since angular momentum ($L = I\omega$) must be conserved, an increase in I forces the angular velocity (ω) to decrease significantly, helping them control the spin and come to a stop.

7) If the ice on the polar caps on the earth melts, how will it affect the duration of the day? Explain.

→ If polar ice melts, the water redistributes toward the equator. This increases the mass distribution away from the axis of rotation, thereby increasing the Earth's moment of inertia (I). Because angular momentum is conserved, an increase in I results in a decrease in angular velocity (ω). Since the duration of a day is inversely proportional to angular velocity ($T = 2\pi/\omega$), the duration of the day will increase (the day becomes slightly longer).

8) The cap of a bottle can be easily opened with the help of two fingers than with one finger. Why?

→ Opening a cap with two fingers utilizes the principle of torque ($\tau = r \times F$). By using two fingers on opposite sides of the cap, you create a couple - two equal and opposite forces acting at a distance from the centre of rotation. This produces a net torque that rotates the cap effectively, whereas using one finger makes it difficult to apply a balanced, rotational force without the finger slipping or applying uneven pressure.

Exam pattern Questions:

1. b) A playground merry-go-round of radius 2.0 m has a moment of inertia 250 kg m^2 and is rotating at 10 rev/min. A 25 kg child jumps onto the edge of the merry-go-round. What is the new angular speed of the merry-go-round?

$$\hookrightarrow r = 2 \text{ m}$$

$$I = 250 \text{ kg m}^2$$

$$f_1 = 10 \text{ rev/min} = \frac{10}{60} \text{ rev/s} = 0.1667 \text{ rev/s}$$

$$\omega_1 = 2\pi f = 2\pi \times \frac{1}{3} = \frac{1}{3} \pi$$

$$m = 25 \text{ kg}$$

Now,

$$I_{\text{child}} = mr^2$$

$$= 25 \times 2^2$$

$$= 100 \text{ kg m}^2$$

$$I_{\text{final}} = I + I_{\text{child}} = 250 + 100 = 350 \text{ kg m}^2$$

Again,

$$I_{\text{initial}} \omega_1 = I_{\text{final}} \omega_2$$

$$\Rightarrow 250 \times \frac{1}{3} \pi = 350 \times \omega_2$$

$$\Rightarrow \omega_2 = \frac{25}{35 \times 3}$$

$$\therefore \omega_2 = 0.75 \text{ rad/s}$$

2.c) A physics teacher stands on a freely rotating platform. He holds a dumbbell in each hand of his outstretched arms while a student gives him a push until his angular velocity reaches 1.5 rad/s . When the freely spinning teacher pulls his hands in close to his body, his angular velocity increases to 5.0 rad/s . What is the ratio of his final kinetic energy to initial kinetic energy?

→ $\omega_0 = 1.5 \text{ rad/s}$

$\omega_1 = 5.0 \text{ rad/s}$

Now,

$$\frac{K.E_1}{K.E_0} = \frac{\frac{1}{2} I_1 \omega_1^2}{\frac{1}{2} I_0 \omega_0^2}$$

$$\Rightarrow \frac{K.E_1}{K.E_0} = \frac{L \omega_1}{L \omega_0} \quad [\because L = I \omega]$$

$$\Rightarrow \frac{K.E_1}{K.E_0} = \frac{5.0}{1.5} \approx 3.33$$

3.

a) A flywheel of moment of inertia 0.32 kg/m^2 is rotated steadily at 120 rad/s by a 50 W electric motor. Find: ~~kin~~

Kinetic energy and angular momentum of the flywheel.

- Value of frictional couple opposing the rotation.
- time taken for the wheel to come to rest after motor has been switched off.

→ Soln,

$$I = 0.32 \text{ kg/m}^2$$

$$P = 50 \text{ W}$$

$$\omega = 120 \text{ rad/s}$$

$$\begin{aligned} K.E &= \frac{1}{2} I \omega^2 \\ &= \frac{1}{2} \times 0.32 \times 120^2 \\ &= 2304 \text{ J} \end{aligned}$$

$$L = I\omega = 0.32 \times 120 = 38.4 \text{ kg m}^2/\text{s}$$

i) the driving torque from the motor perfectly balances the opposing frictional couple (τ):

$$\text{So, } P = \omega \tau$$

$$\Rightarrow \tau = \frac{P}{\omega} = \frac{50}{120} = 0.41 \text{ Nm}$$

$$\text{ii) } \omega_0 = 120 \text{ rad/s}$$

$$\omega = 0 \text{ rad/s}$$

$$\alpha = \frac{\tau}{I} = \frac{0.41}{0.32} \approx 1.302 \text{ rad/s}^2$$

Now,

$$\alpha = \frac{\omega - \omega_0}{t} \Rightarrow t = \frac{-120}{1.302} = 92.16 \text{ s}$$

4.b) A wheel of moment of inertia 20 kg/m^2 about its axis is rotated from rest about its center by a constant torque ' τ ' and energy gained in 10s is 360 J. Calculate:

i) angular velocity at the end of 10s.

ii) No. of rev made by wheel coming to rest if external torque ' τ ' is removed at 10s and a constant opposing torque of 4 Nm is then applied to wheel.

→ Soln,

$$I = 20 \text{ kg/m}^2$$

$$t = 10 \text{ s, } K.E = 360 \text{ J}$$

i) $K.E = \frac{1}{2} I \omega^2$

$\Rightarrow 360 \times 2 = 20 \times \omega^2$

$\Rightarrow \omega^2 = 36$

$\therefore \omega = 6 \text{ rad/s}$

ii) $T = 4 \text{ Nm}$

$\omega_2^2 = \omega_1^2 + 2\alpha\theta$

$\Rightarrow T = 4$

$\Rightarrow 0^2 = 6^2 + 2 \times (-0.2) \cdot \theta$

$\Rightarrow -20\alpha = 4$

$\Rightarrow 0.4\theta = 36$

$\Rightarrow \alpha = -0.2 \text{ rad/s}^2$

$\therefore \theta = 90 \text{ rad}$

Now,

$$n = \frac{\theta}{2\pi} = \frac{90}{2 \times 3.14} \approx 14.32 \text{ rev}$$

5a) A man stands on rotating platform, with his arms stretched horizontally holding a 5 kg weight in each hand. Angular speed is 30 rev/min. The man brings his arms close to his body with the distance of each weight from the axis changing from 90 cm to 20 cm. The moment of inertia of man together with platform may be taken to be constant and equal to 7.6 kg/m^2 .

→ Soln,

$m = 5 \text{ kg}$

$f_0 = \frac{30}{60} \text{ rev/sec} = 0.5 \text{ rev/sec}$

$\omega_0 = \pi \text{ rad/s}$

$I = 7.6 \text{ kg/m}^2$

i) New angular speed ?

ii) Is the kinetic energy conserved in the process? Explain.

iii) What is the ratio of initial K.E of rotation to the final K.E of rotation of system.

Soln,

$$i) I_1 = I_{\text{man+platform}} + 2(mr_1^2) \\ = 7.6 + 2 \times 5 \times (0.9)^2 \\ = 15.7 \text{ kgm}^2$$

Again,

$$I_2 = I_{\text{man+platform}} + 2(mr_2^2) \\ = 7.6 + 2 \times 5 \times (0.2)^2 \\ = 8.0 \text{ kgm}^2$$

We know,

$$T_1 N_1 = I_1 N_1$$

$$\Rightarrow 15.7 \times 30 = 8.0 \times N_2$$

$$\Rightarrow N_2 = 59 \text{ rev/min}$$

ii) No, rotational kinetic energy is not conserved. As the man pulls heavy weight inward he must do internal mechanical work against the outward centrifugal force. This causes the system's kinetic energy to rise.

iii) Soln,

$$\frac{K.E_2}{K.E_1} = \frac{\frac{1}{2} I_2 \omega_2^2}{\frac{1}{2} I_1 \omega_1^2} \\ = \frac{I_2 N_2^2}{I_1 N_1^2}$$

$$\frac{K.E_2}{K.E_1} = \frac{N_2}{N_1} = \frac{59}{30}$$

$$= 1.97$$